

ACE

CYBER R&D Program

Conversational AI Interface for
Analyst-Driven Threat Investigation

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1.0 Introduction

Modern Security Operations Centers (SOCs) are tasked with monitoring, investigating, and responding to increasingly complex cyber threats across large-scale enterprise environments. These environments generate vast volumes of telemetry data in the form of logs, events, and alerts sourced from authentication systems, network devices, endpoints, applications, and cloud infrastructure. While platforms such as Elasticsearch provide powerful indexing and search capabilities for this data, the effectiveness of these tools is fundamentally constrained by how efficiently analysts can query and interpret the underlying datasets.

At present, SOC analysts must translate investigative intent into precise queries, often expressed in Elasticsearch's JSON Domain-Specific Language (DSL). This translation step introduces friction into the investigative workflow, and creates barriers for less-experienced analysts. As log volumes grow and threat-hunting tasks become more time-sensitive, these inefficiencies directly impact detection speed, analyst productivity, and overall defensive posture.

This research explores whether a Conversational AI Interface (CAI) can serve as an effective intermediary between analysts and Elasticsearch, allowing analysts to express threat-hunting intent in natural language while preserving the precision and safety required for operational query execution.

1.1 The Problem

Although modern SIEM platforms offer flexible and expressive query languages, they impose a steep learning curve on SOC analysts. Elasticsearch's JSON DSL, in particular, requires detailed knowledge of field hierarchies, boolean logic, query structure, and time-range semantics. Even routine investigative tasks, such as identifying failed login attempts, correlating suspicious IP activity, or tracing lateral movement, often require analysts to manually construct complex, multi-clause queries.

This manual query construction introduces several operational challenges:

- **Inefficiency:** Writing, testing, and refining queries consumes valuable analyst time, particularly when small syntax errors or incorrect field usage result in failed or misleading queries.
- **Training Bottlenecks:** New analysts require significant onboarding time to become proficient in Elasticsearch query languages, creating scalability issues for growing SOC teams.

- **Inconsistency:** Equivalent investigative questions may be expressed through different queries across analysts, leading to inconsistent results and reduced reproducibility.

As a result, the effectiveness of SOC tooling is limited not by data availability or storage capability, but by the human effort required to interact with that data. This gap between analyst intent and query execution represents a critical inefficiency in current SOC workflows.

1.2 The Motivation

The motivation for this research is to reduce friction between human analysts and large-scale log analysis systems by introducing a controlled, conversational interface that translates natural-language investigative intent into structured Elasticsearch queries. Rather than replacing existing SIEM capabilities, the goal is to augment them by lowering the barrier to effective querying while maintaining strict constraints on correctness, safety, and reproducibility.

From an operational perspective, enabling analysts to ask questions such as “show me failed login attempts in the past 24 hours” or “identify lateral movement between hosts” without manually constructing JSON DSL has the potential to:

- **Accelerate Investigations:** Analysts can move directly from hypothesis to results, reducing time-to-detection.
- **Lower Training Costs:** Junior analysts can perform meaningful threat-hunting tasks without deep prior expertise in query languages.
- **Standardize Analysis:** Natural-language queries can be consistently translated into canonical query structures, improving reproducibility and auditability.
- **Reduce Cognitive Load:** Analysts can focus on interpreting results rather than constructing syntax.

From a research standpoint, this project is motivated by the need to rigorously evaluate whether such an interface can achieve the precision and recall required for SOC use. Rather than relying on subjective usability claims, this study emphasizes quantitative evaluation against deterministic ground truth, using controlled synthetic datasets and objective metrics such as precision, recall, and F1 score.

By addressing both the human and technical dimensions of SOC query workflows, this work aims to assess whether Conversational AI can meaningfully enhance analyst-driven threat investigation without compromising accuracy, safety, or operational discipline.

2.0 Research Question and Hypothesis

The central objective of this research is to evaluate whether a Conversational AI Interface (CAI) can effectively bridge the gap between analyst intent and structured log-query execution within a Security Operations Center (SOC) environment. Rather than assessing user satisfaction or subjective usability, this study focuses on whether natural-language inputs can be translated into accurate, safe, and operationally valid Elasticsearch JSON DSL queries that reliably retrieve threat-relevant events.

2.1 Research Question

How can a Conversational AI interface specifically integrated with Elasticsearch be designed to enable SOC analysts to query log data using natural language for common adversarial threat-hunting tasks (e.g., failed logins, suspicious IPs, lateral movements)?

2.2 Hypothesis

If a Conversational AI interface is developed and integrated with Elasticsearch to support natural-language threat-hunting queries, then, when evaluated against a pre-tagged dataset containing failed-login and lateral-movement events, it will achieve an overall F1 score of at least 0.95 across all evaluated threat-hunting tasks, indicating balanced precision and recall in detecting threat-tagged log events.

2.3 Evaluation Implications

Acceptance or rejection of this hypothesis provides direct insight into the feasibility of deploying Conversational AI as a query interface in SOC contexts. Meeting or exceeding the F1 threshold would indicate that a CAI can reliably support analyst-driven investigations for well-defined threat patterns. Conversely, failure to meet the benchmark would highlight structural limitations in current AI-driven query translation approaches and motivate targeted improvements in prompt design, schema awareness, and query constraint enforcement.

Together, the research question and hypothesis establish a clear, measurable foundation for evaluating whether Conversational AI can meaningfully enhance SOC threat-hunting workflows without sacrificing accuracy or operational rigor.

3.0 System Overview

The system evaluated in this research is a Conversational AI Interface (CAI) deployed within a controlled, cloud-based SOC testbed designed specifically for experimental

evaluation. The CAI functions as an interface-layer augmentation to an Elasticsearch-based log analytics environment, allowing analysts to prompt threat-hunting queries using natural language rather than manually constructing Elasticsearch JSON DSL.

The system is intentionally made as a non-intrusive, read-only component within the broader SOC workflow. All core SIEM capabilities are provided by an Elastic Cloud Enterprise (ECE) deployment hosted within the testbed. The CAI does not alter data, detection rules, or alerting logic; instead, it focuses on translating analyst intent into structured queries that can be safely executed against existing indices.

The testbed itself is designed to be reproducible, auditable, and security-conscious, with clear separation of roles across components such as log generation, data ingestion, query execution, and analyst interaction. This architectural separation allows the CAI's performance to be evaluated independently of infrastructure variability or downstream SOC tooling behavior, as documented in the Testbed Setup & Configuration Plan.

3.1 Conversational AI Interface Concept

Conceptually, the Conversational AI Interface operates as a natural-language-to-query translation layer positioned between the analyst and Elasticsearch. Its purpose is to convert investigative questions into valid, constrained Elasticsearch JSON DSL queries that reflect the analyst's intent while adhering to predefined safety and schema rules.

The CAI follows a structured operational flow:

- **Natural-Language Input**
  An analyst submits a free-form investigative query (e.g., failed logins, suspicious IP activity, lateral movement) through a web-based interface. This interaction simulates how analysts would normally articulate hypotheses during threat-hunting activities.
- **Intent Interpretation and Constraint Enforcement**
  The CAI uses a locally hosted large language model to interpret the semantic intent of the query. This interpretation is bounded by a rule-guided prompt template and schema constraints derived from the known Elasticsearch Common Schema (ECS) fields present in the dataset. These constraints prevent the generation of unsafe, overly broad, or syntactically invalid queries.
- **Structured Query Generation**
  Based on the interpreted intent, the CAI generates exactly one Elasticsearch JSON DSL query. The output is restricted to read-only query constructs, explicit time-

range filtering, bounded result sizes, and known ECS-aligned fields. Queries that violate these constraints are rejected or replaced with clarification requests.

- **Execution and Result Retrieval**
  The generated query is executed against Elasticsearch within the testbed. Returned log events are presented to the analyst without summarization or automated interpretation. The analyst remains fully responsible for reviewing results and making investigative judgments.

This design preserves analyst authority while reducing the cognitive burden associated with manual query construction. Importantly, the CAI does not learn from query results, modify its behavior based on execution feedback, or autonomously refine queries. This constraint ensures that all observed performance reflects the model's initial reasoning and prompt guidance rather than iterative correction.

3.2 Operational Scope and Constraints

The CAI is deliberately scoped to support controlled experimental evaluation rather than full SOC automation. Its operational boundaries are clearly defined within the testbed configuration to ensure safety, reproducibility, and interpretability of results.

In-Scope Capabilities

- Translation of natural-language threat-hunting queries into Elasticsearch JSON DSL
- Support for representative SOC investigative tasks, including:
  - Failed authentication attempts
  - Suspicious or adversarial IP activity
  - Multi-event lateral movement indicators
- Read-only query execution against Elasticsearch indices populated with synthetic, ECS-compliant logs
- Comprehensive auditing of CAI-generated queries and Elasticsearch responses for evaluation purposes

Out-of-Scope Capabilities

- Automated alert generation, triage, or remediation
- Modification of indexed data, detection rules, or SIEM configuration
- Adaptive learning or prompt refinement based on execution outcomes
- Comparison against human SOC analyst performance
- Use of real operational, classified, or sensitive telemetry

Testbed Constraints

- **Schema Dependence:** All queries must reference known ECS fields present in the synthetic dataset. Queries involving nonexistent fields are invalid by design.
- **Synthetic Data Only:** All evaluation is performed using deterministically labeled synthetic logs, enabling objective measurement but limiting environmental realism.
- **Safety Guardrails:** Ambiguous or underspecified inputs must trigger clarification or safe failure rather than speculative query expansion.

4.0 Data Strategy and Ground-Truth Design

Accurate evaluation of a Conversational AI Interface (CAI) for SOC threat-hunting requires a dataset that supports objective measurement, repeatability, and precise ground-truth comparison. In operational SOC environments, real-world log data is obviously not objectively labeled, which makes it very difficult to determine accurate F1 scoring of the CAI's performance. To address these constraints, this research employs a fully synthetic, deterministically labeled dataset generated within a controlled testbed environment.

4.1 Synthetic Data Rationale

All log data used in this study is synthetically generated using custom Python scripts executed within the research testbed. This approach provides several critical advantages for experimental evaluation:

- **Deterministic Ground Truth:** Every generated event is labeled at creation time using explicit rule-based logic, eliminating ambiguity in what constitutes a "correct" result.
- **Controlled Threat Injection:** Adversarial behaviors such as brute-force authentication attempts and lateral movement chains can be introduced at known frequencies and time windows.
- **Reproducibility:** The dataset can be regenerated using the same scripts, parameters, and configurations, enabling repeatable experimentation.
- **Security and Ethics Compliance:** No real PII, CUI, or operational telemetry is used, removing regulatory and ethical constraints associated with real SOC data.

The synthetic dataset is not used for model training. Instead, it serves exclusively as an evaluation corpus against which CAI-generated queries are measured, as documented in the Data Acquisition Report.

4.2 Log Generation and ECS Compliance

Synthetic SOC logs are generated by a dedicated log-generation node within the testbed. The generator produces both benign operational activity and adversarial threat events,

including failed login attempts, suspicious IP access, and lateral movement behaviors. Events are written in NDJSON format and ingested into Elasticsearch via Filebeat on a scheduled basis, simulating continuous log flow.

All generated logs adhere to the Elasticsearch Common Schema (ECS) to ensure structural consistency and query compatibility. Each event includes standardized fields such as timestamps, host identifiers, source and destination IPs, user attributes, event categories, actions, and severity indicators. ECS compliance ensures that CAI-generated queries can rely on stable field names and hierarchies, reducing ambiguity during query construction.

Logs are indexed into date-partitioned Elasticsearch indices (e.g., soc-logs-YYYY.MM.DD), mirroring common SIEM deployment practices and supporting realistic time-range filtering during investigations.

4.3 Ground-Truth Labeling and PI-Key Design

Ground truth in this study is established through a deterministic labeling mechanism embedded directly into the log-generation process. Each generated event is assigned a globally unique identifier (UUID), and its semantic classification (e.g., failed login, lateral movement, benign activity) is recorded in a separate reference file known as the PI-Key.

The PI-Key serves as the authoritative mapping between event UUIDs and their true labels. Because labeling logic is derived directly from the same rule definitions used to generate adversarial behavior, the PI-Key represents an exact, error-free ground truth rather than a probabilistic or manually curated annotation.

For each threat category evaluated in the study, a canonical reference query is also defined. These reference queries implement the same criteria used to tag events during generation and are validated by confirming that their returned UUID sets exactly match the corresponding PI-Key entries. This design enables direct, auditable comparison between CAI-generated queries and expected results.

4.4 Ground-Truth Labeling and PI-Key Design

This data strategy enables precise computation of standard SOC performance metrics, including precision, recall, and F1 score. For each CAI-generated query:

- Returned event UUIDs are extracted from Elasticsearch query results.
- These UUIDs are compared against the PI-Key to compute true positives, false positives, and false negatives.

- Metrics are calculated deterministically, with no reliance on subjective interpretation or heuristic scoring.

Because all logs, labels, and reference queries are generated and stored within the same controlled environment, the evaluation process is fully auditable and repeatable. Any observed errors—such as missed events or over-selection—can be attributed directly to limitations in the CAI's intent interpretation or query construction logic, rather than uncertainty in the dataset itself.

4.5 Limitations of the Data Strategy

While the synthetic dataset enables rigorous evaluation, it also introduces known limitations. Generated logs, although ECS-compliant and structurally realistic, do not fully capture the complexity, variability, or long-term behavioral patterns of real enterprise environments. User activity, host relationships, and adversarial behaviors follow deterministic patterns defined by the generator rather than evolving organically over time.

These limitations are acknowledged and accepted as tradeoffs necessary to achieve objective, quantitative evaluation at an early stage of system maturity. The dataset is therefore well-suited for proof-of-concept validation and comparative testing, but not for claiming production-ready detection performance.

5.0 Experimental Methodology

The experimental methodology for this study is designed to evaluate the accuracy, robustness, and safety of the Conversational AI Interface (CAI) when translating natural-language threat-hunting queries into executable Elasticsearch JSON DSL. All experiments are conducted within a controlled AWS-based SOC testbed, using a fully synthetic, deterministically labeled dataset to enable objective and repeatable measurement.

The methodology follows an applied experimental research design, in which the CAI is treated as an independent variable and its outputs are evaluated against known ground truth. Rather than benchmarking system throughput or infrastructure performance, the focus of this study is on query correctness and retrieval accuracy, reflecting the CAI's intended role as an interface-layer augmentation for SOC analysts.

All experiments are executed according to predefined test cases, with consistent entry criteria, controlled inputs, and standardized scoring procedures.

5.1 Evaluation Metrics

Evaluation metrics are selected to reflect the operational tradeoffs faced by SOC analysts when conducting threat-hunting activities. In this context, both missed threats and excessive false positives impose costs.

Primary Metrics

The primary performance metric used in this study is the F1 score, defined as the harmonic mean of precision and recall:

- Precision measures the proportion of retrieved events that are true threats.
- Recall measures the proportion of true threats that are successfully retrieved.
- F1 Score captures the balance between these two measures.

F1 score is emphasized because it penalizes both under-selection (false negatives) and over-selection (false positives), making it well-suited for SOC evaluation where neither extreme is acceptable.

Supporting Metrics

In addition to F1 score, the following metrics are recorded to support detailed analysis:

- True Positives (TP): Correctly retrieved threat-tagged events
- False Positives (FP): Benign events incorrectly retrieved
- False Negatives (FN): Threat events not retrieved

All metrics are computed by comparing Elasticsearch query results to the PI-Key ground-truth mapping described in Section 4.0. Because the dataset is fully labeled and deterministic, metric computation is exact and reproducible.

5.2 Test Case Overview

Evaluation is conducted through a series of structured test cases designed to assess the CAI across representative SOC threat-hunting tasks. Each test case corresponds to a common investigative scenario and is executed under consistent environmental conditions.

- **Test Case Categories**
  The primary test cases executed in this study include:
- **Baseline Measurement:**
  Establishes idle system behavior and confirms testbed stability prior to CAI interaction.
- **Failed Authentication Detection:**
  Evaluates the CAI's ability to translate natural-language queries related to failed login attempts into accurate Elasticsearch JSON DSL queries.
- **Suspicious IP Activity:**

Tests whether the CAI can correctly generate queries that identify events associated with adversarial or foreign IP addresses within specified time windows.

- **Lateral Movement Detection:**
  Assesses the CAI's performance on more complex, multi-event threat patterns involving host-to-host movement across protocols such as SSH, RDP, SMB, or similar mechanisms.
- **Invalid or Ambiguous Query Handling:**
  Verifies that the CAI responds safely to underspecified or malformed analyst inputs by requesting clarification or declining to generate a query.
- **Linguistic Robustness:**
  Measures whether semantically equivalent natural-language phrasings produce consistent query behavior and retrieval results.

Execution Consistency

For each test case:

- Entry criteria ensure that all services are running and system load is stable.
- The analyst input is issued exactly as defined in the test case.
- The CAI-generated JSON DSL is captured in its raw form.
- Elasticsearch results are collected and evaluated against ground truth.
- Precision, recall, and F1 score are computed using standardized procedures.

Test cases are designed to progressively increase in complexity, allowing the study to identify both strengths and limitations of the CAI across simple and multi-stage investigative tasks.

6.0 Results

This section presents the quantitative results obtained from evaluating the Conversational AI Interface (CAI) across representative SOC threat-hunting tasks. Results are reported using precision, recall, and F1 score metrics computed by comparing CAI-generated query outputs against deterministic ground truth defined by the PI-Key mapping. All results are derived from controlled test executions performed within the research testbed.

Rather than reporting individual run logs, this section focuses on aggregate performance trends, task-level outcomes, and observed strengths and failure modes relevant to SOC operations.

6.1 Overall Performance Summary

Across all evaluated test cases, the CAI demonstrated strong performance on single-event and well-defined threat patterns, particularly failed authentication detection and linguistic robustness tests. Performance degraded on multi-stage behavioral patterns, most notably lateral movement detection, where the CAI exhibited systematic over-selection.

Overall observations:

- High recall was consistently achieved across all threat categories.
- Precision varied significantly depending on task complexity.
- Errors were dominated by false positives rather than false negatives.

6.2 Failed Login Detection

Failed authentication detection consistently produced the strongest results across all evaluations. When analysts issued natural-language queries related to failed logins (e.g., "failed login attempts," "authentication failures"), the CAI generated syntactically valid and well-constrained Elasticsearch JSON DSL queries that closely matched the canonical ground-truth queries.

Key observations:

- Precision remained high, indicating minimal inclusion of benign events.
- Recall approached unity in most runs, with only small numbers of missed events.
- F1 scores met or exceeded the project hypothesis threshold (≥ 0.95).

These results demonstrate that the CAI is well-suited for single-event threat patterns with clear semantic intent and stable ECS field usage.

6.3 Lateral Movement Detection

Lateral movement detection represented the most challenging task for the CAI and produced the weakest overall performance. While the CAI successfully retrieved all true lateral movement events (high recall), it consistently over-selected large numbers of unrelated events.

Key observations:

- Recall was effectively 1.0 across runs.
- Precision was significantly reduced due to thousands of false positives.
- Resulting F1 scores fell well below the success threshold for this task.

This failure mode reflects the inherent difficulty of expressing multi-stage, sequence-dependent behaviors using a single static query without explicit correlation logic. The CAI

tended to rely on broad event categories (e.g., authentication or network activity) rather than precise protocol- and sequence-level constraints.

6.4 Foreign IP Activity Detection

Queries related to suspicious or foreign IP activity produced generally strong results, though with slightly increased variance compared to failed-login detection. In most cases, the CAI successfully constrained queries to relevant IP-related fields and time windows.

Key observations:

- Recall remained high, indicating that most relevant events were retrieved.
- Precision was moderately reduced due to inclusion of marginally related events.
- Occasional schema-related errors required prompt refinement but did not dominate results.

Overall, this category demonstrated that the CAI can handle moderately complex filtering logic when the underlying intent maps cleanly to ECS fields.

6.5 Linguistic Robustness

The CAI exhibited strong robustness to natural-language variation. Semantically equivalent queries phrased using different wording, synonyms, or sentence structures resulted in nearly identical query logic and retrieval behavior.

Key observations:

- All paraphrased queries produced valid JSON DSL.
- Returned UUID sets were highly consistent across phrasing variations.
- F1 scores remained stable and above the target threshold.

This indicates that the CAI effectively captures semantic intent rather than surface-level phrasing, an important property for real-world analyst interaction.

6.6 Safety and Query Validity Results

Across all test cases, the CAI consistently adhered to defined safety and structural constraints:

- All executed queries were syntactically valid Elasticsearch JSON DSL.
- No unrestricted index scans or unsafe query constructs were generated.
- Ambiguous or underspecified inputs triggered clarification rather than speculative query generation.

These results indicate that the CAI's guardrails were effective, even in cases where retrieval accuracy was limited.

7.0 Analysis and Interpretation

The experimental results documented in PRN-001 demonstrate that the Conversational AI Interface (CAI) exhibits a strong translation of well-defined, single-event threat patterns, while showing some structural limitations and prompt sensitivity when applied to multi-stage behaviors such as lateral movement. This section interprets those findings in direct reference to observed executions rather than theoretical expectations.

7.1 Interpretation of Failed Authentication Results

The CAI's performance on failed authentication detection (TC-002) was consistently strong across multiple executions, including baseline failed-login queries and more constrained variants such as foreign failed logins.

Several factors observed in PRN-001 explain this outcome:

1. **Atomic Event Structure**
   Failed authentication attempts are represented as discrete, independent events in the dataset. As documented in RUN-20251203-002 and RUN-20251208-002, the CAI only needed to apply straightforward filters over authentication-related ECS fields to retrieve relevant events.
2. **Explicit Ground-Truth Alignment**
   Ground-truth labels for failed logins are directly tied to the generator rules. This allowed small deviations (e.g., 19 or 33 false negatives) to be clearly identified without ambiguity.
3. **Observed Failure Modes Were Structural, Not Conceptual**
   In later executions, errors arose from:
   - Incorrect use of non-keyword fields (e.g., country_iso_code vs. country_iso_code.keyword)
   - Violations of explicit range-placement rules (placing range outside bool.filter)

   These errors were not misinterpretations of analyst intent, but rather failures to adhere to the strict Elasticsearch DSL constraints. Once corrected through a re-issuance of the same query, the issue was fixed.

These observations support the conclusion that the CAI is effective at semantic intent recognition for simple threat categories, but unfortunately is sensitive to schema-level correctness requirements.

7.2 Linguistic Robustness and Semantic Consistency

The linguistic robustness test (TC-006) provides strong evidence that the CAI interprets semantic intent than a surface level phrasing.
Across five paraphrased failed-login queries in RUN-20251207-006, the CAI generated identical JSON DSL outputs. This behavior was achieved without execution feedback, reinforcing that robustness stems from prompt design and language understanding rather than iterative correction. From a SOC perspective, this indicates that the CAI can accommodate natural variation in analyst phrasing for routine investigative tasks.

7.3 Linguistic Robustness and Semantic Consistency

Lateral movement detection (TC-004) exposed the most significant limitations of the CAI.

In RUN-20251203-004, the CAI:

- Generated a syntactically valid query
- Retrieved all ground-truth lateral movement events (FN = 0)
- Produced 5,068 false positives, resulting in an F1 score of 0.5858

This failure mode was dominated by over-generalization, not missed threats. As noted in the notebook, the CAI relied on broad authentication and network activity filters rather than precise, multi-field correlations.

In later executions (RUN-20251208-004), lateral movement detection improved when:

- Prompts were phrased to closely mirror examples already present in the CAI system prompt
- Additional constraints (e.g., foreign IP filtering) were applied

Under these conditions:

- Precision improved
- False positives were reduced

These outcomes demonstrate that:

- The CAI does not internally reason about attacker progression, only about field-level filtering

Improved performance in later runs reflects prompt engineering success, not a resolved architectural limitation.

7.4 Linguistic Robustness and Semantic Consistency

Across all executions, including failure cases:

- No unsafe queries were executed
- Ambiguous inputs resulted in clarification requests (TC-005)
- No unrestricted index scans or prohibited DSL constructs were observed

This confirms that safety behavior is stable and independent of detection accuracy.

8.0 Limitations

While the results demonstrate that the Conversational AI Interface (CAI) can effectively translate natural-language queries into Elasticsearch JSON DSL for well-defined threat-hunting tasks, the study contains several limitations. These limitations are inherent to the dataset design, system architecture, and experimental scope.

8.1 Synthetic Data Realism

All experiments were conducted using fully synthetic, deterministically labeled SOC logs. Although these logs are ECS-compliant and structurally representative, they do not fully capture the behavioral complexity of real enterprise environments.
Specifically:

- Hostnames, usernames, and IP addresses are randomly combined rather than reflecting stable user–host relationships.
- Event timing follows controlled generation patterns rather than organic, bursty, or irregular behavior.
- Adversarial activity does not evolve dynamically in response to defender actions or environmental changes.

As noted in PRN-001, these constraints limit the dataset's ability to model long-term behavioral baselines, noisy benign activity, and complex attacker dwell-time patterns. The dataset is therefore suitable for controlled accuracy evaluation, but not for claiming operational detection realism.

8.2 Static Query Expressiveness

The CAI is limited to generating a single static Elasticsearch JSON DSL query per analyst input. This constraint significantly impacts performance on threat behaviors that are inherently multi-stage or sequence-dependent.
Observed consequences include:

- Over-selection in lateral movement detection due to reliance on broad categorical filters.
- Inability to encode temporal ordering, causality, or host-to-host relationships.
- Dependence on prompt specificity to approximate correlation behavior.

These limitations are architectural rather than implementation errors. Static DSL queries lack native constructs for expressing attacker progression, making certain SOC use cases fundamentally out of scope for the current system design.

8.3 Prompt Sensitivity and Schema Dependence

PRN-001 documents multiple instances where CAI performance degraded due to schema-level errors, including:

- Incorrect field names or ECS hierarchy placement
- Misuse of .keyword fields
- Violations of explicit range-placement rules

While these issues were correctable through prompt refinement or even simply re-prompting the CAI, they highlight the CAI's dependence on carefully engineered system prompts.

8.4 Absence of Execution Feedback Loop

The CAI does not receive structured feedback from Elasticsearch regarding query success, partial failure, or result quality. Once a query is generated, it is either executed or rejected without informing the model how or why improvements might be made.

This limitation:

- Prevents self-correction of schema or logic errors
- Requires manual prompt adjustment to resolve failures
- Limits adaptability to evolving datasets or index mappings

9.0 Technology Readiness Level Assessment

The Conversational AI Interface (CAI) evaluated in this study is assessed at Technology Readiness Level 4 (TRL-4). This requires that a system's core components be integrated and validated in a controlled environment, with measurable performance against defined criteria. The CAI satisfies these conditions in the following ways:

The CAI is not a standalone algorithm, but a fully integrated prototype consisting of:

- A natural-language interface (Streamlit frontend)
- A locally hosted large language model (via Ollama)
- A rule-guided prompt and validation layer

Live integration with Elasticsearch within an Elastic Cloud Enterprise (ECE) deployment

Validation was performed under laboratory conditions using:

- Fully synthetic, ECS-compliant log data
- Deterministic ground-truth labeling via the PI-Key
- Predefined test cases (TC-001 through TC-006)
- Objective scoring using precision, recall, and F1 score

Performance was evaluated using standardized SOC metrics rather than subjective usability claims. Key observations include:

- High F1 scores for single-event threat patterns (failed logins)
- Consistent semantic interpretation across paraphrased queries

- Documented failure modes for multi-stage behaviors (lateral movement), dominated by false positives rather than missed detections

9.1 Why the System Does Not Meet TRL-5

TRL-5 requires validation in a relevant operational environment, which this study explicitly does not claim.

The CAI does not meet TRL-5 criteria due to the following documented constraints:

- Evaluation is limited to synthetic data rather than live or historical SOC telemetry
- No comparison against human SOC analyst performance
- No integration with production SIEM workflows or alerting pipelines
- No adaptive learning, execution feedback loop, or autonomous refinement
- Performance on complex, correlation-heavy threats remains prompt-dependent and inconsistent

These limitations are intentional and clearly documented in PRN-001 and the Test Plan, and they preclude claims of operational readiness.

9.2 Path to Higher TRL Levels

Advancement beyond TRL-4 would require targeted system and evaluation enhancements, including but not limited to:

- **TRL-5:**
  Validation using realistic SOC datasets (sanitized historical logs or red-team exercises), expanded threat behaviors, and analyst-in-the-loop evaluation.
- **TRL-6:**
  Prototype deployment in a simulated operational SOC environment with real analysts performing live investigations, measuring time-to-detection and cognitive workload reduction.
- **TRL-7+:**
  Integration with production SIEM workflows, continuous feedback loops, and sustained performance under operational load.

These steps represent future work rather than outcomes of the present study.

10.0 Mission Impact and Future Work

This research demonstrates that a Conversational AI Interface (CAI) can reduce friction in analyst-driven threat investigation by enabling natural-language interaction with Elasticsearch while maintaining strict query safety and evaluation rigor. Although the system is not yet operationally deployable, the results establish clear value and a defined path forward.

10.1 Mission Impact

The primary mission impact of this work is the reduction of cognitive and technical barriers associated with SOC log investigation. Experimental results show that, for well-defined threat-hunting tasks such as failed authentication detection, the CAI reliably translates analyst intent into accurate and executable Elasticsearch JSON DSL queries.
This capability has direct relevance to DoD SOC operations by:

- Reducing time and mental overhead required to construct complex queries
- Enabling more consistent and reproducible investigations across analysts
- Supporting junior analysts by lowering the expertise required to perform meaningful threat hunting

Importantly, the CAI functions strictly as an interface-layer augmentation and consistently fails safely when accuracy degrades, preserving analyst authority and minimizing operational risk.

10.2 Future Work

Advancing this capability beyond its current TRL-4 maturity will require targeted development and expanded evaluation. Key future efforts include:

- Improving schema awareness and validation to reduce prompt sensitivity and query errors
- Extending support for multi-stage and correlation-heavy threat behaviors such as lateral movement
- Incorporating analyst-in-the-loop evaluation to measure time savings and usability
- Validating performance on realistic or historical SOC datasets under controlled conditions

These efforts are necessary to transition the CAI from laboratory validation toward operational relevance.

11.0 Conclusion

This research evaluated the feasibility of using a Conversational AI Interface (CAI) to support analyst-driven threat investigation in a Security Operations Center by translating natural-language queries into structured Elasticsearch JSON DSL. Through controlled experimentation in a reproducible SOC testbed and objective comparison against deterministic ground truth, the study demonstrated that conversational interfaces can accurately and safely support well-defined threat-hunting tasks.

Results showed that the CAI consistently achieved high accuracy for single-event investigative patterns such as failed authentication detection and remained robust to natural-language variation. At the same time, the evaluation showed some limitations for complex, multi-stage behaviors, where static query generation and schema sensitivity constrained performance.

Taken together, these findings validate the CAI concept at Technology Readiness Level 4, demonstrating component-level feasibility and measurable benefit under laboratory conditions. While the system is not yet suitable for operational deployment, it establishes a strong foundation for future work focused on schema awareness, correlation-aware querying, and analyst-in-the-loop evaluation.

Overall, this work contributes evidence that conversational interfaces can serve as a practical augmentation to existing SOC tooling, reducing investigative friction while preserving accuracy, safety, and analyst authority.